



BHABHA UNIVERSITY ,BHOPAL

Scheme of Examination

Seventh Semester-Bachelor of Technology

(Electrical and Electronics Engineering) Session 2020-21

SPECIALIZATION- Electrical and Electronics Engineering

S.No.	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory Slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem. Exam.	Mid sem Tests	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ assignment/Quiz /Presentation	
1	EX-71	Microprocessor & Microcontroller	3	1	2	5	70	20	10	30	20	150
2	EX-72	Elective-I	4	1		5	70	20	10			100
3	EX-73	Elective-II	4	1		5	70	20	10			100
4	EX-74	Electrical Cad Lab	-	-	4	2	-	-	-	50	50	100
7	EX-75	Major Project-I	-	-	8	4	-	-	-	100	50	150
8	EX-76	Industrial Training	-	-	6	3	-	-	-	100	50	100
		Total	11	03	20	24	210	60	30	280	170	750

<u>Elective-I</u>	<u>Elective-II</u>
EX-72(A) Electronic Instrumentation	EX-73(A) Digital Signal Processing
EX-72(B) Analog & Digital Communication	EX-73(B) Energy Audit & Management
EX-72(C) Electromagnetic Waves	EX-73(C) Hybrid & Electrical Vehicles

Theory Credits: 12
Practical Credits: 12
Total Credits: 24

Theory Marks: 210
Practical Marks: 280
Quiz, Mid Term, Lab. Work: 260

Total: 750

BHABHAUNIVERSITY,BHOPAL

COURSECURRICULUM FACULTYOFENGINEERING

UndergraduateProgramme

(CourseseffectivefromAcademicYear2020-21)

SYLLABUSOFCOURSESTOBE OFFERED “BachelorofTechnology”

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**Bachelor of Technology VII
Semester
Electrical & Electronics
Engineering (EEE)**

NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462 026

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Bachelor of Technology

Branch-Electrical&ElectronicsEngineering

EX-71 Microprocessor&Micro-controller

UNIT-I

Microprocessor 8086 Introduction to 16-bit 8086 microprocessors, architecture of 8086, Pin Configuration, mode, timing diagram, Memory interfacing, interrupts, Instruction set of 8086, Addressing mode, Assembler directives & operations, assembly and machine language programming, subroutine call and returns, Concept of stack, Stack structure of 8086, timings and delays.

UNIT-II

Input-Output interfacing: Memory Mapped I/O and Peripheral I/O. PPI 8255 Architecture and modes of operation, Interfacing to 16-bit microprocessor and programming, DMA controller (8257) Architecture, Programmable interval timer 8254, USART 8251.

UNIT-III

Microcontroller 8051 Intel family of 8-bit microcontrollers, Architecture of 8051, Pin description, I/O configuration, interrupts; Interrupt structure and interrupt priorities, Port structure and operation, Accessing internal & external memories and different mode of operations, Memory organization, Addressing mode, instruction set of 8051 and programming.

UNIT-IV

8051 Interfacing, Applications and serial communication 8051 interfacing to ADC and DAC, Stepper motor interfacing, Timer/ counter functions, 8051 based thyristor firing circuit, 8051 connections to RS-232, 8051 Serial communication, Serial communication modes, Serial communication programming, Serial port programming in C.

UNIT-V

Microcontroller 8096 Introduction to 16-bit Microcontroller, functional block-diagram, memory status, complete 8096 instruction set, classification of instruction set, addressing modes, programming examples using 8096, hardware features of 8096, parallel ports, control & status Registers, Introduction to 16/32-bit PIC microcontrollers and DSPIC.

REFERENCE BOOKS:-

1. Hall Douglas V., Microprocessor and interfacing, Revised second edition 2006, Macmillan, McGraw Hill.
2. A.K. Ray & K.M. Bhurchandi, Advanced Microprocessors and peripherals - Architecture, Programming and Interfacing, Tata McGraw - Hill, 2009 TMH reprint..
3. Kenneth J. Ayala, The 8086 microprocessor: programming and interfacing the PC, Indian -edition, CENGAGE Learning.
4. Muhammad Ali Mazidi and Janice Gillespie Mazidi, The 8051 Microcontroller and Embedded Systems, Pearson education, 2005.
5. McKinlay, The 8051 Microcontroller and Embedded Systems - using assembly and C, PHI, 2006/ Pearson, 2006.

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EXPERIMENT LIST:-

1. Programs for 16-bit arithmetic operations for 8086 (using various addressing modes).
2. Program for sorting an array for 8086.
3. Program for searching for a number or character in a string for 8086.
4. Program for string manipulations for 8086.
5. Program for digital clock design using 8086.
6. Interfacing ADC and DAC to 8086.
7. Parallel communication between two microprocessors using 8255.
8. Serial communication between two microprocessor kits using 8251.
9. Interfacing to 8086 and programming to control a stepper motor.
10. Programming using arithmetic, logical and bit manipulation instructions of 8051.
11. Program and verify Timer/Counter in 8051.
12. Program and verify Interrupt handling in 8051.

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ELECTIVE-I

EX-72(A) Electronic Instrumentation

UNIT-I

Introduction to CRO, Different parts of CRO, Its Block diagram, Electrostatic focusing, Electrostatic deflection, post deflection acceleration, Screen for CRTs, Graticule, Vertical & Horizontal deflection system, Time base circuit, Oscilloscope probes and transducers, Attenuators, Application of CROs, Lissajous patterns, Special purpose CROs- Multi input, Dual trace, Dual beam, Sampling, Storage (Analog & Digital) Oscilloscopes.

UNIT-II

A.C. Bridge Measurement Sources and detectors, Use of Bridges for measurement of inductance, Capacitance & Q factor Maxwell's bridge, Maxwell's inductance capacitance bridge, Hay's bridge, Anderson's bridge, Owen's Bridge, De-sauty's Bridge, Schering Bridge, High Voltage Schering bridge, Measurement of relative permittivity, Heaviside Campbell's bridge, Weins bridge, Universal bridge, Sources of errors in Bridge circuit, Wagner's Earthing device, Q meter and its applications and measurement methods.

UNIT-III

Transducers Transducers definition and classification, mechanical devices as primary detectors, Characteristic & choice of Transducers, Resistive inductive and capacitive transducers, strain gauge and gauge factor, Thermistor, Thermocouples, LVDT, RVDT, Synchros, Piezo Electric transducers, Magnetoelastic and magnetostrictive Hall effect transducers, Optoelectronic transducers such as photo voltaic, Photo conductive, photo diode and photo conductive cells, Photo transistors, Photo optic transducers. Introduction to analog & Digital data acquisition systems-Instrumentation systems used, Interfacing transducers to electronic control & measuring systems Multiplexing - D/A multiplexing A-D Multiplexing, Special encoders. Digital control description.

UNIT-IV

Signal Generators Fixed & variable frequency AF oscillators, Sine wave generators, Standard signal generator, AF Sine and Square wave generator Function generator, Square and pulse generator, Random noise generator, Sweep generator, TV Sweep generator, Marker generator, Sweep- Marker generator, Wobblyscope, Video pattern generator Vectroscope, Beat frequency oscillator Wave analyser Basic wave analyzer, Frequency selective wave analyzer, Heterodyne wave analyzer, Harmonic distortion, analyzer, spectrum analyzer digital Fourier analyzer.

UNIT-V

Digital Instruments Advantages of Digital instruments over analog instruments, resolution and sensitivity of Digital meters., Digital Voltmeter - Ramp type, Dual slope integration type, Integrating type, Successive approximation type, Continuous balance DVM or Servo balancing potentiometer type DVM. , comparison of Electronic & Digital Volt meter, Digital Multimeter, Digital frequency meter, Time period measurement, High frequency measurement, Electronic counter, Digital tachometer, Digital PH meter, Digital phase meter, Digital capacitance meter. Digital display system and indicators like CRT, LED, LCD, Nixies, Electro luminescent, Incandescent, Electrophoretic image display, Liquid vapour display dot-matrix display, Analog recorders, X-Y recorders. Instruments used in computer- controlled instrumentation RS 232C and IEEE 488, GPIB electric interface.

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REFERENCE BOOKS:-

1. Albert.D.Helfrick,W.D.Cooper,Modern Electronic Instrumentation and measurement techniques, PHI.
2. Kalsi H.S., Electronic Instrumentation, TMH.
3. A.K.Sawhney, Electrical and Electronic measurements and Instrumentation, Dhanpat Rai and Co.
4. E.W.Golding, Electrical Measurement and Measuring Instruments Sir Isaac Pitman and Sons, Ltd. London 19405.
- C.S.Rangan,G.R.Sarma,V.S.V.Mani, Instrumentation Devices and Systems Tata McGraw-Hill Publishing Company Ltd.
6. B.C.Nakra,K.K.Choudhry, Instrumentation, Measurement and Analysis Tata McGraw Hill Publishing Company Ltd.
7. Morris A.S., Principles of Measurement & Instrumentation, PHI

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EX-72(B) Analog & Digital Communication

UNIT-I

Time domain and frequency domain representation of signal, Fourier Transform and its properties, Transform of Gate, Periodic gate, Impulse periodic impulse sine and cosine wave, Concept of energy density and power density (Parseval's theorem), Power density of periodic gate and impulse function, impulse response of a system, convolutions, convolution with impulse function, causal and non-causal system impulse response of ideal low pass filter, Correlation & Auto correlation.

UNIT-II

Base band signal, need of modulation, Introduction of modulations techniques, Amplitude modulation, Equation and its frequency domain representation, Bandwidth, Power distribution. AM suppressed carrier waveform equation and frequency domain representation Generation (Balance/Chopper modulator) and synchronous detection technique, errors in synchronous detection, Introduction to SSB and VSB Transmission Angle modulation, Frequency and phase modulation equation and their relative phase and frequency deviations, modulation index frequency spectrum, NBFM and WBFM, Bandwidth comparison of modulation techniques.

UNIT-III

Sampling of signal, sampling theorem for low pass and Band pass signal, Pulse amplitude modulation (PAM), Time division, multiplexing (TDM). Channel Bandwidth for PAM-TDM signal Type of sampling instantaneous, Natural and flat top, Aperture effect, Introduction to pulse position and pulse duration modulations, Digital signal, Quantization, Quantization error, Pulse code modulation, signal to noise ratio, Companding, Data rate and Baud rate, Bit rate, multiplexed PCM signal, Differential PCM (DPCM), Delta Modulation (DM) and Adaptive Delta Modulation (ADM), comparison of various systems.

UNIT-IV

Digital modulations techniques, Generation, detection, equation and Bandwidth of amplitude shift keying (ASK) Binary Phase Shift Keying (BPSK), Differential phase shift keying (DPSK), offset and non offset quadrature phase shift keying (QPSK), M-Ary PSK, Binary frequency Shift Keying (BFSK), M-Ary FSK Quadrature Amplitude modulation (QAM), MODEM, Introduction to probability of error.

UNIT-V

Information theory and coding- Information, entropies (Marginal and conditional), Model of a communication system, Mathematical representation of source, channel and receiver characteristics, Mutual information, channel capacity efficiency of noise free channel Binary symmetric channel (BSC) Binary erasure channel (BEC), Repetition of signal, NM symmetric Binary channel, Shannon theorem, Shannon-Hartley theorem (S/N-BW trade off) Source encoding code properties; Shannon, Fano and Huffman coding methods and their efficiency error control coding, Minimum Hamming distance, Linear Block Code, Cyclic code and convolution codes. Line Encoding: Manchester coding, RZ, NRZ coding

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REFERENCE BOOKS:-

1. Singh & Sapre, Communication System, TMH
2. Taub & Shilling, Communication System, TMH
3. Hsu; Analog and digital communication (Schaum); TMH
4. B.P. Lathi, Modern Digital and analog communications system,
5. Simon Haykins, Communication System. John Wiley
6. Wayne Tomasi, Electronic Communications system.
7. Martin S. Roden, Analog & Digital Communication System; Discovery Press.
8. Frank R. Dungan, Electronic Communication System, Thomson/Vikas.

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EX-72(C) Electromagnetic Waves

UNIT-I

Transmission Lines Introduction, Concept of distributed elements, Equations of voltage and current, Standing waves and impedance transformation, Lossless and low-loss transmission lines, Power transfer on a transmission line, Analysis of transmission line in terms of admittances, Transmission line calculations with the help of Smith chart, Applications of transmission line, Impedance matching using transmission lines.

UNIT-II

Maxwell's Equations Basic quantities of Electromagnetics, Basic laws of Electromagnetics: Gauss's law, Ampere's Circuit law, Faraday's law of Electromagnetic induction. Maxwell's equations, Surface charge and surface current, Boundary conditions at media interface.

UNIT-III

Uniform Plane Wave Homogeneous unbound medium, Wave equation for time harmonic fields, Solution of the wave equation, Uniform plane wave, Wave polarization, Wave propagation in conducting medium, Phase velocity of a wave, Power flow and Poynting vector.

UNIT-IV

Plane Waves at Media Interface Plane wave in arbitrary direction, Plane wave at dielectric interface, Reflection and refraction of waves at dielectric interface, Total internal reflection, Wave polarization at media interface, Brewster angle, Fields and power flow at media interface, Lossy media interface, Reflection from conducting boundary.

UNIT-V

Waveguides Parallel plane waveguide: Transverse Electric (TE) mode, transverse Magnetic (TM) mode, Cut-off frequency, Phase velocity and dispersion. Transverse Electromagnetic (TEM) mode, Analysis of waveguide-general approach, Rectangular waveguides.

UNIT-VI

Antennas Radiation parameters of antenna, Potential functions, Solution for potential functions, Radiations from Hertz dipole, Near field, Far field, Total power radiated by a dipole, Radiation resistance and radiation pattern of Hertz dipole, Hertz dipole in receiving mode.

REFERENCE BOOKS:-

1. R.K. Shevgaonkar, "Electromagnetic Waves", Tata McGraw Hill, 2005.
2. D.K. Cheng, "Field and Wave Electromagnetics", Addison-Wesley, 1989.
3. M.N.O. Sadiku, "Elements of Electromagnetics", Oxford University Press, 2007.
4. C.A. Balanis, "Advanced Engineering Electromagnetics", John Wiley & Sons, 2012.
5. C.A. Balanis, "Antenna Theory: Analysis and Design", John Wiley & Sons, 2005.

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ELECTIVE-II

EX-73(A) Digital Signal Processing

Unit-I

Discrete-time signals and systems Discrete time signals and systems: Sequences; representation of signals on orthogonal basis; Representation of discrete systems using difference equations, Sampling and reconstruction of signals - aliasing; Sampling theorem and Nyquist rate.

Unit-II

Z-transform z-Transform, Region of Convergence, Analysis of Linear Shift Invariant systems using z-transform, Properties of Z-transform for causal signals, Interpretation of stability in z-domain, Inverse z-transforms.

Unit-III

Discrete Fourier Transform Frequency Domain Analysis, Discrete Fourier Transform (DFT), Properties of DFT, Convolution of signals, Fast Fourier Transform Algorithm, Parseval's Identity, Implementation of Discrete Time Systems.

Unit-IV

Design of Digital filters Design of FIR Digital filters: Window method, Park-McClellan's method. Design of IIR Digital Filters: Butterworth, Chebyshev and Elliptic Approximations; Low-pass, Band-pass, Band stop and High-pass filters. Effect of finite register length in FIR filter design. Parametric and non-parametric spectral estimation. Introduction to multi-rate signal processing.

Unit-V

Applications of Digital Signal Processing Correlation Functions and Power Spectra, Stationary Processes, Optimal filtering using ARMA Model, Linear Mean-Square Estimation, Wiener Filter.

REFERENCE BOOKS:-

1. S.K.Mitra, "Digital Signal Processing: A computer based approach", McGraw Hill, 2011.
2. A.V.Oppenheim and R.W.Schafer, "Discrete Time Signal Processing", Prentice Hall, 1989.
3. J.G.Proakis and D.G.Manolakis, "Digital Signal Processing: Principles, Algorithms And Applications", Prentice Hall, 1997.
4. L.R.Rabiner and B.Gold, "Theory and Application of Digital Signal Processing", Prentice Hall, 1992.
5. J.R.Johnson, "Introduction to Digital Signal Processing", Prentice Hall, 1992.
6. D.J.DeFatta, J.G.Lucas and W.S.Hodgkiss, "Digital Signal Processing", John Wiley & Sons, 1988.

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EX-73(B) Energy Audit & Management

Unit-I

Introduction to energy & power scenario of world, National Energy consumption data, environmental aspects associated with energy utilization; Energy Auditing- need, types, methodology and barriers, role of energy managers, instruments of energy auditing.

Unit-II

Components of EB billing, HT and LT supply, transformers, cable sizing; Concept of capacitors, power factor improvement, harmonics; Electric motors- motor efficiency computation, energy efficient motors; Illumination Lux, Lumens, types of lighting, efficacy, LED lighting and scope of energy conservation in lighting.

Unit-III

Thermal systems, Boilers, Furnaces and Thermic Fluid heaters- efficiency computation and energy conservation measures; Steam distribution and usage, steam traps, condensate recovery, flash steam utilization; Insulation & Refractories.

Unit-IV

Energy conservation in major utilities; pumps, fans, blowers, compressed air systems, Refrigeration & Air Conditioning systems, Cooling Towers, DG sets.

Unit-V

Energy Economics- discount period, payback period, internal rate of return, net present value; Life Cycle costing ESCO concept.

REFERENCE BOOKS:-

1. Witte L.C., Schmidt P.S. and Brown D.R., Industrial Energy Management and Utilization, Hemisphere Publ., Washington, 1988.
2. Callaghan P.W., Design and Management for Energy Conservation, Pergamon Press, Oxford, 1981.
3. Murphy W.R. and McKay G., Energy Management, Butterworths, London, 1987.
4. Energy Manager Training Manual, Bureau of Energy Efficiency (BEE) under Ministry of Power, GOI, 2004

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EX-73(C) Electrical Hybrid Vehicles

UNIT-I

Introduction Conventional Vehicles: Basics of vehicle performance, vehicle power source characterization, transmission characteristics, mathematical model to describe vehicle performance. Introduction to Hybrid Electric Vehicles: History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies. Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis.

UNIT-II

Electric Trains Electric Drive-trains: Basic concept of electric traction, introduction to various electric drive train topologies, power flow control in electric drive-train topologies, fuel efficiency analysis. Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency. drive-train topologies, fuel efficiency analysis.

UNIT-III

Energy Management Strategies Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies.

REFERENCE BOOKS:-

1. C. Mi, M.A. Masrur and D.W. Gao, "Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives", John Wiley & Sons, 2011.
2. S. Onori, L. Serrao and G. Rizzoni, "Hybrid Electric Vehicles: Energy Management Strategies", Springer, 2015.
3. M. Ehsani, Y. Gao, S.E. Gay and A. Emadi, "Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design", CRC Press, 2004.
4. T. Denton, "Electric and Hybrid Vehicles", Routledge, 2016.

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